

A research extension of the Thue–Morse Atlas

New Deductions from Thue–Morse Cancellation

Global spline corrections, exact dyadic reconstruction,
local-limit expansions, and spectral moment twins

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Abstract

The factorization of a Thue–Morse block into binary differences connects integer cancellation, compactly supported smooth functions, and lacunary Fourier products. We exploit the geometry of the derivative copies, rather than only the formal product expansion, to obtain four groups of results. First, finite uniform-convolution splines admit a global, all-orders C^k correction expansion with explicit error bounds and exact leading supremum constants; no neighborhoods of their knots must be removed. Second, at a fixed dyadic argument the finite spline values become an *exact polynomial* in 4^{-m} . A finite Richardson formula therefore recovers the limiting value exactly, with absolute error amplification less than two at every extrapolation order. Third, the positive coefficients obtained by repeatedly summing a finite Thue–Morse block satisfy an exact differential conversion formula, a uniform local-limit expansion with explicit coefficient polynomials, and an exact terminating version on dyadic lattice subsequences. Finally, the normalized Fourier-energy density has moments $2^{r(r+1)}$ and consequently shares every polynomial moment with both a symmetric lognormal density and an explicit family of discrete theta measures. A separated geometric extension explains this last phenomenon beyond the binary case.

All central claims are proved. Exact rational computations are supplied separately for reproducibility. The claimed contribution is the collection of deductions and quantitative refinements established here; classical identities are identified, and worldwide publication priority is not asserted.

Status. An independently checkable research manuscript, not a peer-reviewed publication or a Lean formalization. It develops and tests specific extensions of the supplied repository document. The distinction between exact identities, asymptotic statements, and unresolved extensions is maintained throughout.

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1 Research setting and the main contributions

The signed Thue–Morse sequence is

$$\varepsilon_n = (-1)^{s_2(n)}, \quad n \geq 0, \quad (1.1)$$

where $s_2(n)$ is the number of ones in the binary expansion of n . Its apparent irregularity coexists with exact cancellation on every complete binary block. The classical survey of Allouche and Shallit [2] describes the unusually broad reach of this sequence. Here we investigate one particularly productive direction: what quantitative information survives when those finite cancellations are integrated repeatedly and then passed to a smooth limit?

The supplied *Thue–Morse Atlas and Frontiers* [1] already contains the finite-block factorization, the Fabius–Rvachev bridge, spectral correction coefficients, and a weak spline expansion. Its Part II leaves a strong spline-correction statement away from knots as a conjecture. Our first main theorem proves a global fixed-order statement, including the knots, and identifies the exact leading norm constant. The other principal developments concern finite stopping on dyadic arguments, the local behavior of repeated-prefix coefficients, and a moment-theoretic interpretation of Fourier energy.

The underlying compactly supported function is not new. Its probability interpretation, derivative identities, and rational values at dyadic points belong to the established theory, particularly the work of Arias de Reyna [3, 4]. We use these mechanisms explicitly, reprove the required facts, and do not relabel rationality or the binary product identity as discoveries. Recent work on Thue–Morse transforms, for example [7], provides additional context for finite-block identities; the analytic questions treated here are more specific.

1.1 A compact statement of the results

Let $V_1, V_2, \dots$ be independent, uniformly distributed random variables on $[-1, 1]$. Let u_m and u be the densities of

$$Y_m = \sum_{j=1}^m 2^{-j} V_j, \quad Y = \sum_{j=1}^{\infty} 2^{-j} V_j, \quad h = 2^{-m}. \quad (1.2)$$

Thus u is the Rvachev up-function in the normalization $\text{supp } u = [-1, 1]$ and $u(0) = 1$. Write

$$a_j = \frac{\mathbb{E} Y^{2j}}{(2j)!}, \quad K_r = 2^{r(r+1)/2}. \quad (1.3)$$

The four main conclusions are as follows.

Global correction with a sharp constant. For fixed $k, N \geq 0$, and $m \geq k + 2N + 4$,

$$\left\| D^k \left(u - \sum_{j=0}^N a_j 4^{-jm} u_m^{(2j)} \right) \right\|_{\infty} \leq a_{N+1} K_{k+2N+2} 4^{-(N+1)m}. \quad (1.4)$$

After multiplication by $4^{(N+1)m}$, the norm on the left converges to the constant on the right. In particular, $4^m \|u - u_m\|_{\infty} \rightarrow 4/9$. The proof is in [section 5](#).

Exact dyadic reconstruction. If $x = a/2^d \in (-1, 1)$, then, once $m \geq d + 2$,

$$u_m(x) = \sum_{j=0}^{\lfloor d/2 \rfloor} b_j 4^{-jm} u^{(2j)}(x), \quad \sum_{j \geq 0} b_j z^{2j} = \frac{1}{\mathbb{E} e^{zY}}. \quad (1.5)$$

This is an equality, not an asymptotic series. Consequently $\lfloor d/2 \rfloor + 1$ consecutive spline levels determine $u(x)$ exactly. The reconstruction weights have total absolute value less than two. See [section 6](#).

A local-limit expansion with finite stopping. Let $c_m(k)$ be the coefficients of

$$A_m(z) = \frac{\prod_{j=0}^{m-1} (1 - z^{2^j})}{(1 - z)^m} = \prod_{j=0}^{m-1} (1 + z + \cdots + z^{2^j-1}). \quad (1.6)$$

After the centering and scaling in [section 7](#), their lattice density ρ_m satisfies uniformly on its entire lattice

$$\rho_m(x) = u(x) - \left(\frac{m}{6} + \frac{1}{18}\right) 4^{-m} u''(x) + O(m^2 16^{-m}). \quad (1.7)$$

There are explicit polynomial coefficients at every order. At fixed dyadic points, on the odd-level lattice subsequence, the expansion terminates exactly. The sharp uncorrected uniform error constant is $4/3$ when the error is divided by $m4^{-m}$.

Moment twins of Fourier energy. With angular-frequency Fourier transform $Q(t) = \int e^{itx} u(x) dx$, the probability measure

$$d\nu(t) = \frac{|Q(t)|^2 dt}{\int_{\mathbb{R}} |Q(s)|^2 ds} \quad (1.8)$$

has odd moments zero and even moments $2^{r(r+1)}$. It has the same polynomial moments as Se^L , where S is an independent uniform sign and

$$L \sim N\left(\frac{\log 2}{2}, \frac{\log 2}{2}\right). \quad (1.9)$$

An explicit one-parameter family of purely atomic measures has these moments as well. The identification of this spectral measure, rather than the classical indeterminacy of lognormal moments itself, is the point of [section 10](#).

1.2 Scope of the novelty claim

There are two different questions: whether a theorem is proved, and whether it has appeared anywhere before. The proofs below settle the first for the stated results. A targeted literature check supports presenting the global sharp correction theorem, finite dyadic stopping, and the explicit lattice refinements as research contributions relative to the inspected atlas. That check is not an exhaustive priority search. The Fourier-energy identification is presented as an additional potentially novel connection, with the lognormal background explicitly credited to the classical moment literature [\[5\]](#).

Nothing below purports to solve questions about primes, squares, cubes, normality, or all possible Thue–Morse exponential sums. We instead establish a coherent set of complete results in the finite-block, spline, and Fourier directions.

2 From binary cancellation to compactly supported splines

2.1 The finite product and polynomial cancellation

For $m \geq 0$ define

$$P_m(z) = \sum_{n=0}^{2^m-1} \varepsilon_n z^n. \quad (2.1)$$

Choosing whether each binary digit is present gives the classical identity

$$P_m(z) = \prod_{j=0}^{m-1} (1 - z^{2^j}). \quad (2.2)$$

Indeed, a subset $J \subseteq \{0, \dots, m-1\}$ contributes $(-1)^{|J|} z^{\sum_{j \in J} 2^j}$, and every integer in the block has a unique such representation.

Substitution $z = e^t$ gives

$$\sum_{n=0}^{2^m-1} \varepsilon_n e^{nt} = (-1)^m 2^{m(m-1)/2} t^m + O(t^{m+1}), \quad (2.3)$$

so

$$\sum_{n=0}^{2^m-1} \varepsilon_n n^r = \begin{cases} 0, & 0 \leq r < m, \\ (-1)^m m! 2^{m(m-1)/2}, & r = m. \end{cases} \quad (2.4)$$

This elementary Prouhet cancellation is the algebraic source of all compact-support cancellations used below.

2.2 The continuous convolution model

The j th term $2^{-j} V_j$ has density

$$f_j(x) = 2^{j-1} \mathbf{1}_{[-2^{-j}, 2^{-j}]}(x). \quad (2.5)$$

Therefore $u_m = f_1 * \dots * f_m$. Its support is $[-1+h, 1-h]$, and for $m \geq 2$ it is a C^{m-2} piecewise polynomial of degree at most $m-1$.

The infinite random series in (1.2) converges absolutely for every outcome. The distributional tail is an exact scaled copy of the entire series:

$$Y \stackrel{d}{=} Y_m + hY', \quad Y' \text{ independent of } Y_m, \quad \mathcal{L}(Y') = \mathcal{L}(Y). \quad (2.6)$$

Its characteristic function is

$$Q(t) = \prod_{j=1}^{\infty} \text{sinc}(2^{-j}t), \quad \text{sinc } t = \begin{cases} \sin t/t, & t \neq 0, \\ 1, & t = 0. \end{cases} \quad (2.7)$$

Our Fourier convention is

$$Q(t) = \int_{\mathbb{R}} e^{itx} u(x) dx, \quad u(x) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{-itx} Q(t) dt. \quad (2.8)$$

No factor of 2π is hidden inside the definition of sinc.

Lemma 2.1 (Smoothness and a useful Fourier bound). *For every integer $L \geq 1$,*

$$|Q(t)| \leq \min\{1, K_L |t|^{-L}\}, \quad K_L = 2^{L(L+1)/2}. \quad (2.9)$$

The law of Y has an even density $u \in C_c^\infty(\mathbb{R})$, supported on $[-1, 1]$, with $0 \leq u \leq 1$ and $u(0) = 1$.

Proof. For the first L factors, use $|\text{sinc}(2^{-j}t)| \leq 2^j/|t|$; all remaining factors have modulus at most one. This proves (2.9). Consequently $|t|^r Q(t)$ is integrable for every r , by choosing $L > r+1$. Fourier inversion yields a smooth density. Compact support and evenness follow from the random series. Finally write $Y = V_1/2 + R$, where R is independent and supported on $[-1/2, 1/2]$. Convolution with the unit-height density of $V_1/2$ proves $u \leq 1$ and $u(0) = 1$. The same reasoning shows $\|u_m\|_\infty = 1$. $\square$

2.3 A finite signed formula

For $y \in \mathbb{R}$ write $y_+ = \max\{y, 0\}$.

Proposition 2.2 (Thue–Morse positive-part formula). *For $m \geq 2$,*

$$u_m(x) = \frac{2^{m(m-1)/2}}{(m-1)!} \sum_{n=0}^{2^m-1} \varepsilon_n (x+1-h-2hn)_+^{m-1}. \quad (2.10)$$

In particular the knots are among $-1+h+2hn$. For $0 \leq r \leq m-2$, differentiation gives

$$u_m^{(r)}(x) = \frac{2^{m(m-1)/2}}{(m-1-r)!} \sum_{n=0}^{2^m-1} \varepsilon_n (x+1-h-2hn)_+^{m-1-r}. \quad (2.11)$$

The analogous formula for $r = m-1$ holds on open cells.

Proof. The distributional derivative of f_j is $2^{j-1}(\delta_{-2^{-j}} - \delta_{2^{-j}})$. Hence

$$u_m^{(m)} = 2^{m(m-1)/2} \sum_{n=0}^{2^m-1} \varepsilon_n \delta_{-1+h+2hn}. \quad (2.12)$$

To see the locations and signs, select an upper endpoint in any subset of the m factors. The lengths 2^{1-j} , when put in reverse order, are $2h, 4h, \dots, 2^m h$. Their subset sums give $2hn$ and their signs are ε_n . Integrating m times from $-\infty$ proves (2.10). The polynomial terms to the right of the support cancel by (2.4). The derivative formula follows. $\square$

The signed formula is excellent for exact rational computation, but its large cancelling summands make naive floating-point evaluation unreliable. A stability statement for extrapolation is not a stability statement for this inner signed sum.

3 The derivative-copy geometry

The key analytic fact is that, after differentiation, the rescaled copies have disjoint interiors. Thus differentiating a fixed number of times does *not* introduce a factor growing with m .

Proposition 3.1 (Exact derivative copies and norm identities). *For integers $r \geq 0$ and $m \geq r+2$,*

$$u_m^{(r)}(x) = K_r \sum_{n=0}^{2^r-1} \varepsilon_n u_{m-r}(2^r x + 2^r - 1 - 2n), \quad (3.1)$$

$$u^{(r)}(x) = K_r \sum_{n=0}^{2^r-1} \varepsilon_n u(2^r x + 2^r - 1 - 2n). \quad (3.2)$$

For every $1 \leq p \leq \infty$,

$$\left\| u_m^{(r)} \right\|_p = K_r \|u_{m-r}\|_p, \quad \left\| u^{(r)} \right\|_p = K_r \|u\|_p, \quad (3.3)$$

$$\left\| u_m^{(r)} \right\|_\infty = K_r, \quad \left\| u^{(r)} \right\|_\infty = K_r. \quad (3.4)$$

Proof. Differentiating the first box and rescaling the tail gives

$$u'_m(x) = 2\{u_{m-1}(2x+1) - u_{m-1}(2x-1)\}. \quad (3.5)$$

The same argument applies to u . On iterating, the accumulated factor is $2^{1+2+\dots+r} = K_r$. At every step, endpoint choices contribute binary digit signs, proving the two copy identities.

The n th infinite copy is supported on

$$\left[-1 + \frac{2n}{2^r}, -1 + \frac{2n+2}{2^r} \right]. \quad (3.6)$$

These intervals meet only at their endpoints. Finite copies have still smaller supports. Thus there is no cancellation in the integral of the p th power of the absolute value. A change of variable contributes 2^{-r} for each of the 2^r copies, giving (3.3). The supremum statement uses $\|u_j\|_\infty = \|u\|_\infty = 1$. $\square$

Remark 3.2 (What is classical, and what is used here). Derivative self-similarity is an established part of the Fabius theory [3, 4]. We emphasize the exact norm consequence and its uniformity in the finite level. That simple observation is the mechanism behind the global remainder theorem and the spectral-moment calculation below.

Corollary 3.3 (Vanishing high derivatives at a dyadic point). *If $x = a/2^d \in (-1, 1)$ and $r > d$, then $u^{(r)}(x) = 0$.*

Proof. In (3.2), $2^r x$ is an even integer and $2^r - 1 - 2n$ is odd. Every argument is therefore an odd integer. The density u is zero at and outside ± 1 . $\square$

This does not say that u agrees with its Taylor polynomial on a neighborhood of x . Vanishing higher derivatives at one point and local polynomiality are very different assertions for a C^∞ function.

4 Explicit coefficient calculus

The moment generating function

$$M(z) = \mathbb{E}e^{zY} = \prod_{j=1}^{\infty} \frac{\sinh(2^{-j}z)}{2^{-j}z} = \sum_{j=0}^{\infty} a_j z^{2j} \quad (4.1)$$

is entire. Its reciprocal is analytic near the origin. We shall need both its coefficients and those of two closely related functions.

Define

$$\lambda_r = \frac{\zeta(2r)}{r\pi^{2r}}, \quad r \geq 1. \quad (4.2)$$

The sine product [6, Sec. 4.22], or its logarithmic derivative followed by integration, gives

$$-\log \operatorname{sinc} z = \sum_{r \geq 1} \lambda_r z^{2r}, \quad (4.3)$$

$$\log M(z) = \sum_{r \geq 1} \frac{(-1)^{r+1} \lambda_r}{4^r - 1} z^{2r}. \quad (4.4)$$

The first coefficients are $\lambda_1 = 1/6$, $\lambda_2 = 1/180$, and $\lambda_3 = 1/2835$. Euler's even-zeta formula makes every λ_r rational.

For any sequence $v_1, v_2, \dots$, define the finite partition polynomial

$$\mathcal{E}_j(v) = \sum_{\substack{k_1, \dots, k_j \geq 0 \\ \sum_{r=1}^j r k_r = j}} \prod_{r=1}^j \frac{v_r^{k_r}}{k_r!}, \quad \mathcal{E}_0(v) = 1. \quad (4.5)$$

It is characterized by $\exp(\sum_{r \geq 1} v_r z^r) = \sum_{j \geq 0} \mathcal{E}_j(v) z^j$. Thus all coefficients in this article have explicit finite-sum expressions, without requiring auxiliary recurrences.

Proposition 4.1 (The four coefficient families). *Let*

$$\frac{1}{M(z)} = \sum_{j \geq 0} b_j z^{2j}, \quad (4.6)$$

$$\left(\frac{z}{\sinh z} \right)^m = \sum_{j \geq 0} \gamma_j(m) z^{2j}, \quad (4.7)$$

$$\exp \left(\sum_{r \geq 1} \lambda_r \left(m + \frac{1}{4^r - 1} \right) z^r \right) = \sum_{j \geq 0} H_j(m) z^j. \quad (4.8)$$

Then

$$a_j = \mathcal{E}_j \left(\frac{(-1)^{r+1} \lambda_r}{4^r - 1} \right), \quad (4.9)$$

$$b_j = (-1)^j \mathcal{E}_j \left(\frac{\lambda_r}{4^r - 1} \right), \quad (4.10)$$

$$\gamma_j(m) = (-1)^j \mathcal{E}_j(m \lambda_r), \quad (4.11)$$

$$H_j(m) = \mathcal{E}_j \left(\lambda_r \left(m + \frac{1}{4^r - 1} \right) \right). \quad (4.12)$$

For each j , $H_j(m)$ is a rational polynomial of degree j with positive coefficients and leading term $m^j / (6^j j!)$. Moreover,

$$\sum_{\ell=0}^j \gamma_\ell(m) b_{j-\ell} = (-1)^j H_j(m). \quad (4.13)$$

Proof. Expand the exponentials in (4.4)–(4.8). For the alternating families, the sign of each partition monomial is $(-1)^{\sum r k_r} = (-1)^j$. The only degree- j term in m comes from $k_1 = j$. Multiplication of the two reciprocal generating functions gives (4.13). $\square$

For reference, the first a_j and b_j are

j	a_j	b_j
0	1	1
1	1/18	−1/18
2	19/16200	31/16200
3	583/42865200	−2347/42865200
4	132809/1311675120000	1904369/1311675120000

The first lattice correction polynomials are

$$\begin{aligned} H_0(m) &= 1, \\ H_1(m) &= \frac{m}{6} + \frac{1}{18}, \\ H_2(m) &= \frac{(m + 1/3)^2}{72} + \frac{m + 1/15}{180}, \\ H_3(m) &= \frac{(m + 1/3)^3}{1296} + \frac{(m + 1/3)(m + 1/15)}{1080} + \frac{m + 1/63}{2835}. \end{aligned} \quad (4.14)$$

4.1 An optional rational recurrence for computation

Writing $\mu_{2j} = \mathbb{E}Y^{2j}$, one may alternatively use

$$(2j+1)(4^j-1)\mu_{2j} = \sum_{\ell=0}^{j-1} \binom{2j+1}{2\ell} \mu_{2\ell}, \quad \mu_0 = 1. \quad (4.15)$$

For completeness, $M(2z) = M(z) \sinh z/z$. Multiplying by z , comparing the coefficient of z^{2j+1} , and moving the last term to the left proves (4.15). This recurrence is convenient for exact arithmetic, but (4.9) is a nonrecursive alternative.

5 Global spline corrections, sharpness, and increasing order

Define the corrected spline

$$C_{m,N}(x) = \sum_{j=0}^N a_j h^{2j} u_m^{(2j)}(x). \quad (5.1)$$

Theorem 5.1 (Global all-orders correction). *Let $k, N \geq 0$ be integers, put $r = k + 2N + 2$, and suppose $m \geq r + 2$. Then*

$$\|D^k(u - C_{m,N})\|_{\infty} \leq a_{N+1} K_r h^{2N+2}. \quad (5.2)$$

The same right-hand side bounds $\|u - C_{m,N}\|_{C^k} := \max_{0 \leq \ell \leq k} \|D^{\ell}(u - C_{m,N})\|_{\infty}$. The estimate is global on $\mathbb{R}$, including all spline knots and the endpoints of the supports.

Proof. The exact tail identity (2.6) gives

$$u^{(k)}(x) = \mathbb{E} u_m^{(k)}(x - hY). \quad (5.3)$$

Set $g = u_m^{(k)}$. By the hypothesis on m , g has $2N + 2$ continuous derivatives. Taylor's formula with the integral remainder implies, for real y ,

$$\left| g(x - hy) - \sum_{\ell=0}^{2N+1} \frac{(-hy)^{\ell}}{\ell!} g^{(\ell)}(x) \right| \leq \frac{h^{2N+2} |y|^{2N+2}}{(2N+2)!} \|g^{(2N+2)}\|_{\infty}. \quad (5.4)$$

Average over Y . All odd powers disappear because Y is symmetric. The remaining finite sum is precisely $D^k C_{m,N}$, and the remainder is bounded by $a_{N+1} h^{2N+2} \|u_m^{(r)}\|_{\infty}$. Now apply (3.4). For lower derivatives use $K_{\ell+2N+2} \leq K_r$. $\square$

Corollary 5.2 (Global Hölder control). *Let $0 < \alpha \leq 1$, $r = k + 2N + 2$, and $m \geq r + 3$. With $[f]_{\alpha} = \sup_{x \neq y} |f(x) - f(y)|/|x - y|^{\alpha}$,*

$$[D^k(u - C_{m,N})]_{\alpha} \leq 2^{1+\alpha r} a_{N+1} K_r h^{2N+2}. \quad (5.5)$$

Thus a global fixed-order Hölder version is available as well.

Proof. For a bounded differentiable function, the two estimates $|f(x) - f(y)| \leq 2 \|f\|_{\infty}$ and $|f(x) - f(y)| \leq \|f'\|_{\infty} |x - y|$ imply $[f]_{\alpha} \leq (2 \|f\|_{\infty})^{1-\alpha} \|f'\|_{\infty}^{\alpha}$. Apply theorem 5.1 at derivative orders k and $k + 1$, and use $K_{r+1} = 2^{r+1} K_r$. $\square$

Why the knots are harmless here. The differentiation order is fixed before the level m tends to infinity. The spline degree and its classical smoothness grow with m , while the norm of each admissible fixed derivative is exactly constant in m . Thus there is no knot singularity in the derivatives used in the Taylor remainder. A different argument would be required for arbitrary derivative orders beyond the stated regularity threshold.

Theorem 5.3 (Uniform leading remainder and sharp norm constant). *For fixed $k, N \geq 0$,*

$$h^{-2N-2} D^k(u - C_{m,N}) \longrightarrow a_{N+1} u^{(k+2N+2)} \quad \text{uniformly on } \mathbb{R}. \quad (5.6)$$

Consequently

$$\lim_{m \rightarrow \infty} 4^{(N+1)m} \|D^k(u - C_{m,N})\|_{\infty} = a_{N+1} K_{k+2N+2}. \quad (5.7)$$

Proof. Apply [theorem 5.1](#) with order $N + 1$ to obtain

$$D^k(u - C_{m,N}) = a_{N+1} h^{2N+2} u_m^{(k+2N+2)} + O_{k,N}(h^{2N+4}) \quad (5.8)$$

in the supremum norm. The same theorem at correction order zero, with derivative order $k + 2N + 2$, proves convergence of the displayed spline derivative to $u^{(k+2N+2)}$. Uniform convergence implies convergence of the supremum norms, and [proposition 3.1](#) evaluates the limiting norm. $\square$

For the first two un-differentiated cases,

$$\lim_{m \rightarrow \infty} 4^m \|u - u_m\|_{\infty} = \frac{4}{9}, \quad (5.9)$$

$$\lim_{m \rightarrow \infty} 16^m \left\| u - u_m - \frac{4^{-m}}{18} u_m'' \right\|_{\infty} = \frac{2432}{2025}. \quad (5.10)$$

These constants are not estimates fitted to numerical data. They are forced by the exact derivative norms.

5.1 A certified increasing-order construction

The next result allows the correction order to grow with the spline level. It is an absolute error statement, not a relative endpoint asymptotic.

Corollary 5.4 (Supergeometric absolute accuracy). *Fix $k \geq 0$, let $m \geq k + 4$, and choose r to be the largest integer with $r \leq m - 2$ and $r \equiv k \pmod{2}$. Set $N = (r - k - 2)/2$. Then*

$$\|D^k(u - C_{m,N})\|_{\infty} \leq \frac{2^{-m^2/2 + (k+1/2)m+3}}{(m - k - 3)!}. \quad (5.11)$$

Proof. Since $|Y| \leq 1$, $a_{N+1} \leq 1/(r - k)!$. The bound in [theorem 5.1](#) becomes

$$\frac{2^{r(r+1)/2 - (r-k)m}}{(r - k)!}. \quad (5.12)$$

Write $r = m - \delta$, where $\delta \in \{2, 3\}$. The exponent equals $-m^2/2 + (k + 1/2)m + \delta(\delta - 1)/2$, and $(r - k)! \geq (m - k - 3)!$. This proves the claim. $\square$

The large positive and negative terms in an actual evaluation still need controlled arithmetic. The error certificate establishes the quality of the exact corrected expression; it does not provide a complete floating-point or bit-complexity analysis.

6 Exact stabilization on dyadic arguments

Fix $x = a/2^d \in (-1, 1)$. It is harmless to choose a nonminimal d ; a smaller denominator exponent simply gives a sharper degree bound.

Lemma 6.1 (A low-degree cell around a dyadic point). *For $m \geq d + 2$, the restriction of u_m to $[x - h, x + h]$ agrees with one polynomial of degree at most d .*

Proof. The knots have coordinates $-1 + h + 2hn$, which are odd multiples of h when $m \geq 1$. Since $2^m x$ is even, x is the center of an open cell whose endpoints are $x - h$ and $x + h$.

Apply the derivative-copy formula at order $d + 1$ on this open cell. It remains valid there in the borderline case as a piecewise derivative. Put $\ell = m - d - 1 \geq 1$. Each argument of u_ℓ has the form

$$\text{an odd integer} + 2^{d+1}(y - x), \quad |y - x| < h. \quad (6.1)$$

Its distance from zero is larger than $1 - 2^{-\ell}$, unless it is farther away still. But u_ℓ is supported on $[-1 + 2^{-\ell}, 1 - 2^{-\ell}]$. Thus $u_m^{(d+1)} = 0$ on the cell. The spline is a polynomial there, so its degree is at most d . Continuity gives the endpoint equality as well. $\square$

Theorem 6.2 (Exact finite tail calculus at dyadic points). *For $x = a/2^d \in (-1, 1)$, $m \geq d + 2$, and $0 \leq r \leq d$,*

$$u^{(r)}(x) = \sum_{j=0}^{\lfloor (d-r)/2 \rfloor} a_j h^{2j} u_m^{(r+2j)}(x), \quad (6.2)$$

$$u_m^{(r)}(x) = \sum_{j=0}^{\lfloor (d-r)/2 \rfloor} b_j h^{2j} u^{(r+2j)}(x). \quad (6.3)$$

Both formulas are exact.

Proof. The tail formula (5.3) averages the derivative of u_m at points $x - hY \in [x - h, x + h]$. On this cell the derivative is a polynomial of degree at most $d - r$. Its Taylor formula has no remainder. Symmetry gives (6.2).

View the vector of derivatives of orders $r, r + 2, \dots$ at x as a finite vector. The first formula is a triangular transformation with diagonal entries one. The power-series identity $M(z)/M(z) = 1$ shows that its inverse has coefficients b_j , which proves (6.3). This is inversion of a finite triangular system, not an assertion that an infinite differential operator converges on an arbitrary smooth function. $\square$

Corollary 6.3 (Polynomial stabilization). *For fixed $x = a/2^d$, there is a polynomial R_x of degree at most $p = \lfloor d/2 \rfloor$ such that*

$$u_m(x) = R_x(4^{-m}), \quad m \geq d + 2, \quad R_x(0) = u(x). \quad (6.4)$$

Its coefficient of z^j is $b_j u^{(2j)}(x)$.

Corollary 6.4 (The degree is exact in lowest terms). *Suppose $d \geq 1$ and a is odd, so $a/2^d$ is in lowest terms. The polynomial R_x has degree exactly $p = \lfloor d/2 \rfloor$. More explicitly, its highest derivative factor is*

$$u^{(2p)}(a/2^d) = \begin{cases} K_d \varepsilon_{(a+2^d-1)/2}, & d \text{ even}, \\ \frac{1}{2} K_{d-1} \varepsilon_{\lfloor (a+2^d)/4 \rfloor}, & d \text{ odd}. \end{cases} \quad (6.5)$$

Proof. For even d , precisely one argument in the order- d copy formula is zero, and all the others are outside the support. For odd d , at order $d - 1$ precisely one argument is $1/2$ or $-1/2$. The corresponding values are $u(0) = 1$ and $u(\pm 1/2) = 1/2$. The latter equality follows directly by convolving the first box with its symmetric tail. The indices in (6.5) identify these copies. Both values are nonzero. Finally $b_p \neq 0$ by (4.10), so the highest coefficient of R_x is nonzero. $\square$

The crucial word is “polynomial.” An ordinary all-orders asymptotic expansion, by itself, would not justify finite exact reconstruction. Here the geometry of the dyadic cell supplies the missing exactness.

6.1 Finite Richardson reconstruction

For $0 < q < 1$ and $p \geq 0$, let

$$(q; q)_j = \prod_{\ell=1}^j (1 - q^\ell), \quad (q; q)_0 = 1, \quad (6.6)$$

and define

$$w_{p,j}(q) = \prod_{\substack{0 \leq \ell \leq p \\ \ell \neq j}} \frac{-q^\ell}{q^j - q^\ell} = \frac{(-1)^{p-j} q^{(p-j)(p-j+1)/2}}{(q; q)_j (q; q)_{p-j}}. \quad (6.7)$$

These are simply the Lagrange interpolation weights for evaluation at zero from the nodes $1, q, \dots, q^p$.

Theorem 6.5 (Exact reconstruction and uniform stability). *Let $x = a/2^d \in (-1, 1)$, $p = \lfloor d/2 \rfloor$, and $m \geq d + 2$. Then*

$$u(x) = \sum_{j=0}^p w_{p,j}(1/4) u_{m+j}(x). \quad (6.8)$$

At every order,

$$\sum_{j=0}^p |w_{p,j}(1/4)| = \prod_{\ell=1}^p \frac{1 + 4^{-\ell}}{1 - 4^{-\ell}} < 2. \quad (6.9)$$

Thus absolute errors bounded by δ in each input spline value produce an absolute output error smaller than 2δ .

Proof. Apply Lagrange interpolation to the polynomial $z \mapsto R_x(4^{-m}z)$ from corollary 6.3. This proves the equality.

Taking absolute values in (6.7), and using the finite q -binomial product identity, gives

$$\begin{aligned} \sum_{j=0}^p |w_{p,j}(q)| &= \frac{1}{(q; q)_p} \sum_{k=0}^p q^{k(k+1)/2} \frac{(q; q)_p}{(q; q)_k (q; q)_{p-k}} \\ &= \frac{\prod_{\ell=1}^p (1 + q^\ell)}{(q; q)_p}. \end{aligned} \quad (6.10)$$

The finite product identity follows by expanding $\prod_{\ell=1}^p (1 + tq^\ell)$ and comparing the coefficient of t^k ; alternatively its coefficients satisfy the same two-term recurrence on adjoining the last factor.

For the strict bound, $p = 0, 1$ are immediate. At $q = 1/4$, the first two factors multiply to $17/9$. For the remaining factors use

$$\sum_{\ell=3}^{\infty} \log \frac{1 + q^\ell}{1 - q^\ell} \leq \frac{2q^3}{(1 - q)(1 - q^6)} = \frac{512}{12285}. \quad (6.11)$$

Since $e^t \leq (1-t)^{-1}$ for $0 \leq t < 1$, the entire product is at most

$$\frac{17}{9} \frac{1}{1 - 512/12285} = \frac{208845}{105957} < 2. \quad (6.12)$$

Finally apply the triangle inequality to the input perturbations. $\square$

The first three weight rows are

$$(1), \quad \left(-\frac{1}{3}, \frac{4}{3}\right), \quad \left(\frac{1}{45}, -\frac{4}{9}, \frac{64}{45}\right). \quad (6.13)$$

These familiar extrapolation weights are not new in isolation. The point is the exact stopping theorem that makes them recover physical-space dyadic values in finitely many levels.

6.2 Examples and a closed finite formula

Example 6.6. The following identities hold at every indicated sufficiently large level:

$$\begin{aligned} u_m(1/2) &= \frac{1}{2}, & m \geq 3, \\ u_m(1/4) &= \frac{67}{72} + \frac{4}{9} 4^{-m}, & m \geq 4, \\ u_m(1/8) &= \frac{287}{288} + \frac{2}{9} 4^{-m}, & m \geq 5, \\ u_m(1/16) &= \frac{2073457}{2073600} + \frac{5}{162} 4^{-m} - \frac{3968}{2025} 16^{-m}, & m \geq 6. \end{aligned} \quad (6.14)$$

For example, $u''(1/16) = -5/9$ and $u^{(4)}(1/16) = -1024$; substitution into (6.3) gives the last line. The stated thresholds are uniform sufficient thresholds, not claims of minimality for each example.

Combining (6.8) with (2.10) gives a closed finite expression for a rational dyadic value. Put $m = d+2$ and $p = \lfloor d/2 \rfloor$:

$$\begin{aligned} u(x) &= \sum_{j=0}^p \frac{w_{p,j}(1/4) 2^{(m+j)(m+j-1)/2}}{(m+j-1)!} \\ &\quad \times \sum_{n=0}^{2^{m+j}-1} \varepsilon_n \left(x + 1 - 2^{-(m+j)} - 2^{1-m-j} n \right)_+^{m+j-1}. \end{aligned} \quad (6.15)$$

Every quantity on the right is rational. This gives another proof of classical dyadic rationality, and more importantly exhibits a finite extrapolation mechanism with a uniform condition bound. The formula is not advertised as the fastest existing algorithm.

7 Repeated-prefix coefficients: an exact differential identity

7.1 The positive coefficient array and its natural normalization

Let $c_m(k)$ be the coefficient of z^k in $A_m(z)$ from (1.6), extended by zero outside its support. Put

$$D_m = 2^m - m - 1, \quad B_m = A_m(1) = 2^{m(m-1)/2}. \quad (7.1)$$

The support is $0 \leq k \leq D_m$. The coefficients are positive integers on this interval and are symmetric about $D_m/2$.

Repeated inclusive summation multiplies a generating function by $(1 - z)^{-1}$. Therefore these are precisely the m -fold cumulative coefficients of the finite signed block, and

$$c_m(k) = \sum_{n=0}^{\min(k, 2^m-1)} \varepsilon_n \binom{k-n+m-1}{m-1}, \quad k \geq 0. \quad (7.2)$$

For $k < 2^m$ this agrees with the m -fold inclusive prefix of the infinite Thue–Morse sequence. Beyond that range, the finite block and infinite sequence should not be conflated.

Let $D_0^*, \dots, D_{m-1}^*$ be independent random variables with D_j^* uniform on $\{0, \dots, 2^j - 1\}$. Then

$$\mathbb{P} \left(\sum_{j=0}^{m-1} D_j^* = k \right) = \frac{c_m(k)}{B_m}. \quad (7.3)$$

Set

$$\Delta = 2h = 2^{1-m}, \quad x_{m,k} = \Delta(k - D_m/2) = -1 + (m+1)h + 2hk, \quad (7.4)$$

and define the scaled lattice mass

$$\rho_m(x_{m,k}) = \frac{c_m(k)}{\Delta B_m}. \quad (7.5)$$

The division by Δ is essential: ρ_m is a density-scale quantity, not the probability of one lattice point.

7.2 A centered factorial polynomial identity

The next lemma is a useful form of the generalized Bernoulli polynomial calculus; the standard generating function is recorded in [6]. Its short proof is included to fix all normalizations.

Lemma 7.1 (Centered factorial conversion). *For $m \geq 2$ define*

$$p_{m-1}(w) = \prod_{r=1}^{m-1} (w + r - m/2). \quad (7.6)$$

As an identity of polynomials,

$$p_{m-1}(w) = \left(\frac{D_w/2}{\sinh(D_w/2)} \right)^m w^{m-1}. \quad (7.7)$$

The power series on the right acts finitely: every derivative of order greater than $m-1$ annihilates the polynomial.

Proof. By coefficient extraction and the formal change of variable $y = e^z - 1$,

$$\begin{aligned} [z^{m-1}] \left(\frac{z}{2 \sinh(z/2)} \right)^m e^{wz} &= \text{Res}_{z=0} \frac{e^{(w+m/2)z}}{(e^z - 1)^m} dz \\ &= \text{Res}_{y=0} \frac{(1+y)^{w+m/2-1}}{y^m} dy \\ &= \binom{w+m/2-1}{m-1} = \frac{p_{m-1}(w)}{(m-1)!}. \end{aligned} \quad (7.8)$$

The first coefficient equals $1/(m-1)!$ times the differential operator in (7.7) applied to w^{m-1} . The change of variable is valid formally because $e^z - 1 = z + O(z^2)$. $\square$

Theorem 7.2 (Exact lattice-to-spline differential conversion). *For $m \geq 2$ and every $k \in \mathbb{Z}$,*

$$\rho_m(x_{m,k}) = \sum_{j=0}^{\lfloor (m-1)/2 \rfloor} \gamma_j(m) h^{2j} u_m^{(2j)}(x_{m,k}). \quad (7.9)$$

When m is odd the highest derivative is a cellwise derivative, but the lattice nodes are then cell centers. When m is even the highest derivative used is of order $m - 2$ and is continuous. Thus every displayed evaluation is unambiguous.

Proof. At $x = x_{m,k}$ put

$$t = \frac{x + 1 - h}{2h} = k + \frac{m}{2}, \quad J_m = 2^{-(m-1)(m-2)/2}. \quad (7.10)$$

The signed spline formula becomes

$$u_m(x) = \frac{J_m}{(m-1)!} \sum_{\substack{0 \leq n < 2^m \\ n < t}} \varepsilon_n (t - n)^{m-1}. \quad (7.11)$$

On the other hand, (7.2) and $1/(\Delta B_m) = J_m$ give

$$\rho_m(x) = \frac{J_m}{(m-1)!} \sum_{\substack{0 \leq n < 2^m \\ n \leq k}} \varepsilon_n p_{m-1}(t - n). \quad (7.12)$$

One can extend this last sum from $n \leq k$ to $n < t$: every new integer has $n = k + i$ with $1 \leq i < m/2$, and then $p_{m-1}(m/2 - i) = 0$. The same observation works if the original sum is empty or if the finite block truncates the range.

Now apply lemma 7.1 term by term to (7.11). Since $D_t = 2hD_x$, its operator becomes $(hD_x/\sinh(hD_x))^m$, whose coefficients are $\gamma_j(m)h^{2j}$. This proves the formula. At an even-level knot, the derivatives involved have positive remaining polynomial degree; the limiting formula agrees from both sides. $\square$

The first terms are

$$\rho_m(x) = u_m(x) - \frac{m}{6} h^2 u_m''(x) + \left(\frac{m^2}{72} + \frac{m}{180} \right) h^4 u_m^{(4)}(x) + \cdots, \quad (7.13)$$

where the complete expression is the finite sum in (7.9). This identity is stronger than a formal differential approximation.

8 The all-orders uniform local-limit theorem

The exact operator in theorem 7.2 is useful algebraically. To obtain a uniform fixed-order remainder as m grows, it is cleaner to work on the lattice Fourier interval.

8.1 Exact removal of the lattice

Proposition 8.1 (Continuous noise representation). *Let*

$$X_m = \Delta \left(\sum_{j=0}^{m-1} D_j^* - D_m/2 \right). \quad (8.1)$$

There is a random variable W_m , independent of X_m , such that

$$Y \stackrel{d}{=} X_m + hW_m, \quad W_m = V_1 + \cdots + V_m + Y', \quad (8.2)$$

where all summands on the right are independent, the V_j are uniform on $[-1, 1]$, and Y' is a copy of Y . In particular

$$\text{Var}(W_m) = \frac{m}{3} + \frac{1}{9}. \quad (8.3)$$

The characteristic function of X_m satisfies

$$\psi_m(t) = \frac{Q(t)}{Q(ht) \text{sinc}(ht)^m} \quad \left(|t| \leq \frac{\pi}{2h} \right). \quad (8.4)$$

Proof. Adjoin independent U_j uniform on $[0, 1]$ to D_j^* . Then $D_j^* + U_j$ is uniform on $[0, 2^j]$, and

$$\Delta \left(D_j^* + U_j - 2^{j-1} \right) = \Delta \left(D_j^* - \frac{2^j - 1}{2} \right) + h(2U_j - 1). \quad (8.5)$$

Summing over j produces X_m plus m independent uniform noises of half-width h . The continuous sum has the law of Y_m after reversing the order of its widths. Adjoin the tail hY' . The variance follows from $\mu_2 = 1/9$. Taking characteristic functions proves the quotient wherever its denominator is nonzero, in particular on the stated interval. $\square$

This calculation explains the leading factor m . Continuous spline truncation omits one self-similar tail of scale h , whereas converting a lattice sum to its continuous counterpart requires m additional small uniforms. Their combined variance is of order mh^2 .

8.2 Uniform expansion and an explicit remainder constant

For use in the proof, write

$$\mathcal{H}_m(z) = \sum_{j \geq 0} H_j(m) z^j = \exp \left(\sum_{r \geq 1} \lambda_r \left(m + \frac{1}{4^r - 1} \right) z^r \right). \quad (8.6)$$

By (8.4), $\psi_m(t) = Q(t) \mathcal{H}_m(h^2 t^2)$ on the fundamental interval.

Theorem 8.2 (Uniform all-orders local-limit expansion). *For each fixed integer $N \geq 0$, there exists an explicit constant C_N such that for every $m \geq 4$,*

$$\sup_{k \in \mathbb{Z}} \left| \rho_m(x_{m,k}) - \sum_{j=0}^N (-1)^j H_j(m) h^{2j} u^{(2j)}(x_{m,k}) \right| \leq C_N (m+1)^{N+1} h^{2N+2}. \quad (8.7)$$

One permissible, deliberately nonoptimized choice is obtained by putting $L = 4N + 8$ and taking

$$C_N = \frac{2}{\pi} \left\{ \frac{1}{2N+3} + \frac{K_L}{2N+5} + \frac{K_L}{4N+7} + K_L \sum_{j=0}^N \frac{1}{4N+7-2j} \right\}. \quad (8.8)$$

Proof. Lattice Fourier inversion and our normalization give

$$\rho_m(x_{m,k}) = \frac{1}{2\pi} \int_{-\pi/(2h)}^{\pi/(2h)} e^{-itx_{m,k}} \psi_m(t) dt. \quad (8.9)$$

The finite proposed approximation is the full-line inverse Fourier transform of $Q(t) \sum_{j=0}^N H_j(m) h^{2j} t^{2j}$, because multiplication by t^{2j} corresponds to $(-1)^j D^{2j}$. We estimate the difference of these transforms in L^1 ; the resulting bound is independent of k .

Low frequencies. Put $T = h^{-1/2}$ and $R = (m+1)^{-1}$. Since $\lambda_r \leq 2/(r\pi^{2r})$, positivity gives

$$\log \mathcal{H}_m(R) \leq \frac{2}{\pi^2 - R} \leq \frac{2}{\pi^2 - 1} < \log 2. \quad (8.10)$$

Hence

$$\mathcal{H}_m(R) < 2, \quad H_j(m) \leq 2(m+1)^j. \quad (8.11)$$

For $|t| \leq T$, put $z = h^2 t^2 \leq h$. Since $m \geq 4$, $z/R \leq (m+1)h \leq 5/16 < 1$. Positivity of all coefficients implies

$$\left| \mathcal{H}_m(z) - \sum_{j=0}^N H_j(m) z^j \right| \leq 2((m+1)z)^{N+1}. \quad (8.12)$$

Therefore the low-frequency error before division by 2π is at most

$$2(m+1)^{N+1} h^{2N+2} \int_{\mathbb{R}} |t|^{2N+2} |Q(t)| dt. \quad (8.13)$$

Using [lemma 2.1](#) with $L = 4N + 8$ bounds the integral by

$$2 \left(\frac{1}{2N+3} + \frac{K_L}{2N+5} \right). \quad (8.14)$$

High frequencies of the exact lattice transform. For $|s| \leq \pi/2$, $\text{sinc } s \geq 2/\pi$. Also, multiplying the bounds $\text{sinc}(2^{-j}s) \geq 1 - 2^{-2j}s^2/6$ gives

$$Q(s) \geq 1 - \frac{s^2}{18} \geq 1 - \frac{\pi^2}{72} > \frac{1}{2}. \quad (8.15)$$

Thus on the fundamental interval,

$$|\psi_m(t)| \leq 2(\pi/2)^m |Q(t)|. \quad (8.16)$$

Write $\beta = \log_2(\pi/2) < 1$. Integration over $|t| > T$ yields

$$\int_{T < |t| \leq \pi/(2h)} |\psi_m(t)| dt \leq \frac{4K_L}{L-1} h^{(L-1)/2-\beta} \leq \frac{4K_L}{4N+7} h^{2N+2}. \quad (8.17)$$

High frequencies of the finite approximation. By (8.11) and [lemma 2.1](#),

$$\begin{aligned} & \int_{|t| > T} |Q(t)| \sum_{j=0}^N H_j(m) h^{2j} |t|^{2j} dt \\ & \leq 4K_L \sum_{j=0}^N \frac{(m+1)^j h^{(L-1)/2+j}}{L-2j-1} \\ & \leq 4K_L (m+1)^{N+1} h^{2N+2} \sum_{j=0}^N \frac{1}{4N+7-2j}. \end{aligned} \quad (8.18)$$

Add (8.13)–(8.18), divide by 2π , and use $(m+1)^{N+1} \geq 1$ in the exact-transform tail. The result is precisely (8.7) with (8.8). $\square$

Remark 8.3 (Meaning of the local limit). The supremum in (8.7) includes nodes outside the finite support, where $c_m(k) = 0$. This is a uniform comparison of scaled point masses with a density evaluated at the same nodes. It is not convergence in total variation of a discrete measure to an absolutely continuous one.

Corollary 8.4 (Sharp first local-limit error). *Uniformly for $k \in \mathbb{Z}$,*

$$\frac{\rho_m(x_{m,k}) - u(x_{m,k})}{m4^{-m}} + \frac{1}{6}u''(x_{m,k}) \longrightarrow 0. \quad (8.19)$$

In particular,

$$\lim_{m \rightarrow \infty} \frac{\sup_{k \in \mathbb{Z}} |\rho_m(x_{m,k}) - u(x_{m,k})|}{m4^{-m}} = \frac{4}{3}. \quad (8.20)$$

Proof. Use theorem 8.2 with $N = 1$ and $H_1(m)/m \rightarrow 1/6$. Its remainder divided by $m4^{-m}$ tends to zero. The node spacing tends to zero, so the sampled supremum of the continuous compactly supported function $|u''|$ tends to $\|u''\|_\infty = 8$ by proposition 3.1. $\square$

9 Exact dyadic stopping for the integer coefficient array

The local-limit expansion has an unexpectedly exact specialization. It is obtained from the finite operator, not by declaring an asymptotic series to terminate.

Theorem 9.1 (Exact dyadic lattice expansion). *Fix $x = a/2^d \in (-1, 1)$. If m is odd and $m \geq d + 2$, then*

$$k_m = 2^{m-1}(1+x) - \frac{m+1}{2} \in \mathbb{Z}, \quad x_{m,k_m} = x, \quad (9.1)$$

and

$$\rho_m(x) = \sum_{j=0}^{\lfloor d/2 \rfloor} (-1)^j H_j(m) 4^{-jm} u^{(2j)}(x). \quad (9.2)$$

The right-hand side is an exact finite sum of polynomials in m times powers of 4^{-m} .

Proof. The stated parity makes k_m an integer. On the cell about x , u_m has degree at most d by lemma 6.1. Hence the finite differential conversion in theorem 7.2 has no nonzero terms with $2j > d$ at this point. Substitute (6.3) into its remaining terms. The coefficient of $h^{2j} u^{(2j)}(x)$ is the convolution $\sum_{\ell=0}^j \gamma_\ell(m) b_{j-\ell}$, which is $(-1)^j H_j(m)$ by (4.13). $\square$

Example 9.2 (A closed repeated-prefix identity). For every odd $m \geq 5$,

$$\rho_m(1/4) = \frac{67}{72} + \left(\frac{4m}{3} + \frac{4}{9} \right) 4^{-m}. \quad (9.3)$$

Equivalently, an explicitly indexed coefficient of the positive Thue–Morse quotient is

$$c_m \left(5 \cdot 2^{m-3} - \frac{m+1}{2} \right) = 2^{m(m-1)/2-m+1} \left\{ \frac{67}{72} + \left(\frac{4m}{3} + \frac{4}{9} \right) 4^{-m} \right\}. \quad (9.4)$$

This also evaluates the corresponding m -fold inclusive prefix of the infinite sequence, since the index is below 2^m .

Proof. At $x = 1/4$, $u = 67/72$, $u'' = -8$, and all derivatives of order above two vanish. Apply theorem 9.1 and then undo the density normalization. $\square$

At $x = 1/16$ the exact identity has three terms:

$$\rho_m(1/16) = \frac{2073457}{2073600} + \frac{5}{9}H_1(m)4^{-m} - 1024H_2(m)16^{-m}, \quad m \geq 7 \text{ odd.} \quad (9.5)$$

The increasing polynomial factors $H_j(m)$ are therefore not merely artifacts of a Fourier proof; they occur in exact rational coefficient identities.

A useful distinction. For the spline values at a fixed dyadic point, the coefficients of 4^{-jm} are constants depending on that point. For the discrete array values, the coefficients are polynomials in m . The extra polynomials come from the m independent lattice-removal noises. This distinction prevents the incorrect use of the simple spline Richardson weights as an exact annihilator of all lattice errors.

10 Fourier-energy moments and lognormal twins

We now use the L^2 version of the derivative-copy identity. This has a different flavor from either approximation theorem.

Definition 10.1 (The normalized energy measure). Let

$$I_0 = \int_{\mathbb{R}} |Q(t)|^2 dt = 2\pi \|u\|_2^2, \quad d\nu(t) = I_0^{-1} |Q(t)|^2 dt. \quad (10.1)$$

This is an absolutely continuous probability measure on the angular frequency line. It is not the singular diffraction measure of the unscaled Thue–Morse substitution system.

Theorem 10.2 (Exact energy moments). *Every moment of ν exists, and for $r \geq 0$,*

$$\int_{\mathbb{R}} t^{2r} d\nu(t) = 2^{r(r+1)}, \quad \int_{\mathbb{R}} t^{2r+1} d\nu(t) = 0. \quad (10.2)$$

Proof. The decay estimate (2.9) gives existence of every moment. Evenness gives the odd moments. Plancherel’s identity and (3.3) at $p = 2$ give

$$\frac{\int t^{2r} |Q(t)|^2 dt}{\int |Q(t)|^2 dt} = \frac{\|u^{(r)}\|_2^2}{\|u\|_2^2} = K_r^2 = 2^{r(r+1)}. \quad (10.3)$$

□

10.1 An explicit absolutely continuous twin

Theorem 10.3 (Lognormal moment twin). *Let S take values $-1, 1$ with equal probabilities and let $L \sim N((\log 2)/2, (\log 2)/2)$ be independent of S . Then the law of $T = Se^L$ has the same moments as ν , but it is a different probability measure. Its density is*

$$g(t) = \frac{1}{2|t|\sqrt{\pi \log 2}} \exp\left(-\frac{(\log |t| - (\log 2)/2)^2}{\log 2}\right), \quad t \neq 0, \quad g(0) = 0. \quad (10.4)$$

Consequently the Hamburger moment problem for (10.2) is indeterminate.

Proof. For a normal random variable of mean μ and variance σ^2 , completion of the square in its density gives $\mathbb{E}e^{sL} = e^{s\mu + s^2\sigma^2/2}$. At $s = 2r$ and the stated parameters, this is $2^{r(r+1)}$. The independent sign kills all odd moments. Change of variable $t = e^L$, followed by symmetrization, gives (10.4).

The density g extends continuously to zero with value zero, whereas the density of ν has value $1/I_0 > 0$ there. They therefore differ on a neighborhood of zero, not merely at a single point. Thus two distinct measures have the same complete moment sequence. □

Lognormal moment indeterminacy is classical [5]. The contribution here is the exact identification of a compactly supported smooth function’s Fourier-energy measure with that moment class. No inference from the failure of a sufficient uniqueness criterion is needed: two distinct measures have been constructed explicitly.

10.2 A continuum of atomic twins

For $\alpha \in [0, 1)$ set

$$\Theta(\alpha) = \sum_{k \in \mathbb{Z}} 2^{-(k+\alpha)^2}. \quad (10.5)$$

It is finite and strictly positive.

Theorem 10.4 (Theta-supported moment twins). *Every measure*

$$\nu_\alpha = \frac{1}{2\Theta(\alpha)} \sum_{k \in \mathbb{Z}} 2^{-(k+\alpha)^2} (\delta_{2^{k+\alpha+1/2}} + \delta_{-2^{k+\alpha+1/2}}) \quad (10.6)$$

is a probability measure with the moment sequence (10.2).

Proof. The total mass is one and the odd moments vanish. For the even moments, complete the square in the exponent:

$$\begin{aligned} \int t^{2r} d\nu_\alpha(t) &= \frac{1}{\Theta(\alpha)} \sum_{k \in \mathbb{Z}} 2^{-(k+\alpha)^2+2r(k+\alpha+1/2)} \\ &= \frac{2^{r^2+r}}{\Theta(\alpha)} \sum_{k \in \mathbb{Z}} 2^{-(k+\alpha-r)^2} = 2^{r(r+1)}. \end{aligned} \quad (10.7)$$

The last equality shifts the integer summation index by r . All sums converge absolutely. $\square$

The theta construction is a standard type of discrete lognormal moment mechanism; the displayed proof is included rather than claiming that mechanism itself as a new discovery. Its combination with the energy identity gives the exact polynomial quadrature

$$\frac{1}{I_0} \int_{\mathbb{R}} p(t) |Q(t)|^2 dt = \int_{\mathbb{R}} p(t) g(t) dt = \int_{\mathbb{R}} p(t) d\nu_\alpha(t) \quad \text{for every polynomial } p. \quad (10.8)$$

This equality cannot be extended to all bounded continuous test functions, since the measures are distinct.

10.3 Identical jets, and what restores uniqueness

Define the normalized autocorrelation

$$R_u(x) = \frac{\int_{\mathbb{R}} u(y)u(y+x) dy}{\|u\|_2^2}. \quad (10.9)$$

It is a C^∞ function supported on $[-2, 2]$ and is the characteristic function of ν . The energy moments give

$$R_u^{(2r)}(0) = (-1)^r 2^{r(r+1)}, \quad R_u^{(2r+1)}(0) = 0. \quad (10.10)$$

Its formal Taylor series at zero has radius of convergence zero, since $2^{r(r+1)}/(2r)!$ has unbounded $2r$ th root. The characteristic functions of the lognormal and atomic twins have exactly the same derivatives at zero. Thus their differences from R_u are smooth and flat at zero, but not identically zero.

Moments alone lose information, but the original scaling relation retains it.

Proposition 10.5 (Refinement restores uniqueness). *Suppose a probability density f is continuous at zero and satisfies pointwise*

$$f(2t) = \text{sinc}^2(t)f(t), \quad t \in \mathbb{R}. \quad (10.11)$$

Then $f(t) = I_0^{-1}Q(t)^2$.

Proof. Iterating toward zero gives

$$f(t) = f(2^{-n}t) \prod_{j=1}^n \text{sinc}^2(2^{-j}t). \quad (10.12)$$

Continuity at zero and convergence of the product imply $f(t) = f(0)Q(t)^2$. Integration fixes $f(0) = 1/I_0$. $\square$

11 A separated geometric extension

The moment phenomenon is not an accidental equality special to one choice of units. It follows from separation of derivative copies. Here is a one-parameter extension that makes this explicit.

For $0 < \theta \leq 1/2$, let ν_θ be the density of

$$Y_\theta = (1 - \theta) \sum_{j=0}^{\infty} \theta^j V_j. \quad (11.1)$$

It is supported on $[-1, 1]$ and is smooth by the same Fourier argument as before. At $\theta = 1/2$ it equals u .

Theorem 11.1 (Separated-copy norms and their moment class). *For $r \geq 0$ and $1 \leq p \leq \infty$,*

$$\left\| \nu_\theta^{(r)} \right\|_p = (2\theta)^{r/p} [2(1 - \theta)]^{-r} \theta^{-r(r+1)/2} \|\nu_\theta\|_p, \quad (11.2)$$

where $(2\theta)^{r/p} = 1$ at $p = \infty$. If ν_θ is its normalized Fourier-energy measure, then

$$\int t^{2r} d\nu_\theta(t) = [2(1 - \theta)^2]^{-r} \theta^{-r^2}, \quad \int t^{2r+1} d\nu_\theta(t) = 0. \quad (11.3)$$

These are the moments of a symmetric lognormal variable with logarithmic mean and variance

$$\mu_\theta = -\frac{1}{2} \log(2(1 - \theta)^2), \quad \sigma_\theta^2 = -\frac{1}{2} \log \theta. \quad (11.4)$$

Proof. Differentiating the first uniform factor yields

$$\nu'_\theta(x) = \frac{1}{2(1 - \theta)\theta} \left\{ \nu_\theta \left(\frac{x + 1 - \theta}{\theta} \right) - \nu_\theta \left(\frac{x - 1 + \theta}{\theta} \right) \right\}. \quad (11.5)$$

The two copies have disjoint interiors because $\theta \leq 1/2$. At depth r there are 2^r disjoint copies, each of horizontal scale θ^r , with signs given by binary endpoint parity. Their common vertical multiplier is

$$[2(1 - \theta)]^{-r} \theta^{-1-2-\dots-r}. \quad (11.6)$$

Integrating the p th power gives (11.2). At $p = 2$, squaring the norm ratio simplifies to $[2(1 - \theta)^2]^{-r} \theta^{-r^2}$. Plancherel proves the moment formula. The normal exponential-moment calculation gives (11.4). $\square$

For a completely explicit atomic version, put $a = -\log \theta > 0$, $\mu = \mu_\theta$, and

$$\Theta_a(\alpha) = \sum_{k \in \mathbb{Z}} e^{-a(k+\alpha)^2}. \quad (11.7)$$

Then

$$\frac{1}{2\Theta_a(\alpha)} \sum_{k \in \mathbb{Z}} e^{-a(k+\alpha)^2} (\delta_{e^{\mu+a(k+\alpha)}} + \delta_{-e^{\mu+a(k+\alpha)}}) \quad (11.8)$$

has the same moments, by shifting k after completing the square. This generalization concerns separated copies. For $\theta > 1/2$ the copies overlap, the simple norm calculation no longer applies, and (11.3) is not asserted.

12 Exact computations and reproducibility

The accompanying `verify.py` uses only the Python standard library. Algebraic tests use arbitrary-size integers and `Fraction`, not floating-point agreement. The saved `verification_results.json` records the actual run used in preparing this article.

12.1 What was checked

The script performed 3,796 categorized exact checks: 35 Prouhet moment identities, 135 derivative-copy identities, 2,307 dyadic polynomial identities, 127 exact Richardson reconstructions, 510 prefix-convolution identities, 15 extrapolation-norm identities, 501 lattice differential conversions, and 166 dyadic lattice identities. The tested ranges and loop bounds are visible in the script. Some checks compare formulas obtained from the same underlying theory; they are useful for catching sign, indexing, and normalization errors, not substitutes for the general proofs.

The implementation constructs the positive coefficient array by sliding-window convolution with all-ones vectors. At level j the vector length is $O(2^j)$, so all levels through m use $O(2^m)$ integer operations. This is an arithmetic-operation count, not a bit-complexity bound: the integers themselves grow.

12.2 A nondyadic corrected approximation

For $x = 1/3$, the reference value was computed from $C_{18,6}(x)$. The exact global certificate from [theorem 5.1](#) is less than 3.469×10^{-59} , and the reference starts

$$u(1/3) = 0.8198348851985181 \dots \quad (12.1)$$

The rational reference and its rational error bound are included in the JSON output. The entries in [table 1](#) are rounded absolute differences from that reference; its certified uncertainty is negligible at the digits displayed.

For example, at $m = 12$, $N = 2$, the actual displayed error is approximately 4.95223×10^{-21} , while the rigorous global bound is approximately 6.03996×10^{-21} . The same certificate works at every real x , not only at this sample point.

12.3 An exact discrete-array experiment

At $x = 1/4$ and odd levels the normalized error is exactly $4/3 + 4/(9m)$ by (9.3). The script computed the coefficient arrays independently by positive convolution and obtained [table 2](#). In every row the residual after subtracting the two terms of (9.3) is exactly zero.

m	$ u - u_m $	$ u - C_{m,1} $	$ u - C_{m,2} $
6	$8.90547 \cdot 10^{-5}$	$5.96017 \cdot 10^{-8}$	—
8	$5.56024 \cdot 10^{-6}$	$2.29567 \cdot 10^{-10}$	$8.44219 \cdot 10^{-14}$
10	$3.47493 \cdot 10^{-7}$	$8.95576 \cdot 10^{-13}$	$2.03133 \cdot 10^{-17}$
12	$2.17182 \cdot 10^{-8}$	$3.49806 \cdot 10^{-15}$	$4.95223 \cdot 10^{-21}$

Table 1: Corrected spline errors at $x = 1/3$. The omitted entry has not been used because the uniform classical-regularity threshold for $N = 2$ is $m \geq 8$.

m	$\rho_m(1/4)$	$\frac{\rho_m(1/4) - 67/72}{m4^{-m}}$
5	15/16	64/45
7	1907/2048	88/63
9	15247/16384	112/81
11	487881/524288	136/99
13	1951517/2097152	160/117
15	124897055/134217728	184/135

Table 2: Exact repeated-prefix coefficients after density normalization. The last column tends to $4/3$.

The script also makes finite numerical checks of the theta moment formulas at several values of α . Those checks are explicitly numerical; the completion-of-the-square proof, not truncation of a theta sum, establishes the exact identity for every moment.

13 Consequences, limitations, and concrete next problems

The strongest conclusion for the motivating approximation question is that fixed-order strong spline corrections need no deleted knot neighborhoods. The sufficient condition is a regularity threshold on the spline level. This directly addresses the strong-spline conjecture in the supplied atlas [1]; it does not establish that document’s separate relative-error endpoint saddle evaluator.

The exact dyadic results exhibit a second phenomenon. A function can be globally nonanalytic and yet admit finite exact extraction from an approximation sequence at every dyadic point. The source of this exactness is a low-degree *finite spline cell*, not analyticity of the limit. The highest nonzero derivative even has an explicit Thue–Morse sign in (6.5).

On the discrete side, the exact centered-factorial identity separates two sources of correction: $\gamma_j(m)$ accounts for lattice removal, and b_j accounts for the infinite continuous tail. Their convolution produces $H_j(m)$. This is an effective way to organize further coefficient identities without repeatedly rediscovering their signs and centering terms.

Finally, the spectral calculation shows that an entire family of energy moments can conceal major differences between probability measures. A continuous density positive at zero, a lognormal-type density flat at zero, and a purely atomic measure can share every polynomial moment. The refinement equation restores information that the moments alone discard.

Research problem 13.1 (Relative endpoint accuracy). Develop bounds for $C_{m,N}(x) - u(x)$ relative to $u(x)$ as x approaches an endpoint while m and N vary. The present global absolute bounds remain true there, but division by the very small value of $u(x)$ can make them ineffective. A proof will need local

information beyond a global derivative norm.

Research problem 13.2 (Efficient exact extraction). Determine the best bit complexity of dyadic reconstruction from positive coefficient arrays, local spline cells, or the signed formula (6.15). The extrapolation condition bound is uniform, but the cost of producing the inputs and the cost of large integer cancellation must be analyzed separately.

Research problem 13.3 (Overlapping geometric copies). For $1/2 < \theta < 1$, compute or characterize the cross terms in $\left\|v_{\theta}^{(r)}\right\|_2^2$. The separated calculation in theorem 11.1 identifies exactly where the argument changes: the derivative-copy intervals now overlap. It is natural to ask which parameters, if any, retain an explicit moment class.

13.1 A formalization route without a formalization claim

The main dependency chain is short. Finite convolution and binary endpoint enumeration yield the derivative-copy identity. Disjoint supports give its norm formula. A Taylor remainder bound and the probability-tail identity then yield the global theorem. The dyadic cell lemma and finite triangular inversion are algebraic once that bridge is available. Separately, the centered-factorial identity can be formalized in a polynomial ring, and the lattice local-limit theorem requires Fourier inversion with explicit integrable majorants. Plancherel plus disjoint supports proves the energy moments.

These are proposed proof components, not assertions that new Lean theorems have been compiled or added to the repository. The manuscript and the exact-arithmetic script are the deliverables of this investigation.

A Normalization dictionary and implementation details

A.1 Frequency conventions

The angular-frequency transform used here is $Q(t)$ in (2.7). Under the cyclic-frequency convention $\widehat{u}(\xi) = \int e^{-2\pi i \xi x} u(x) dx$, evenness gives

$$\widehat{u}(\xi) = Q(2\pi\xi) = \prod_{j=0}^{\infty} \text{sinc}(\pi\xi/2^j). \quad (\text{A.1})$$

A finite cyclic product indexed $j = 0, \dots, n$ therefore corresponds to the spline u_{n+1} in this article. Its omitted-tail scale is $2^{-(n+1)}$, not 2^{-n} . The even moments of its normalized cyclic-frequency energy measure are

$$(2\pi)^{-2r} 2^{r(r+1)}. \quad (\text{A.2})$$

This dictionary is important when comparing formulas with repository documents that use a different Fourier convention or start their product index at zero.

A.2 Integer-only evaluation of a spline derivative

For $x = a/b$ with $b > 0$ and $p = m - 1 - r \geq 1$, put

$$T = a2^m + b(2^m - 1). \quad (\text{A.3})$$

Then (2.11) is

$$u_m^{(r)}(a/b) = \frac{2^{m(m-1)/2}}{p!(b2^m)^p} \sum_{\substack{0 \leq n < 2^m \\ T-2bn > 0}} \varepsilon_n (T - 2bn)^p. \quad (\text{A.4})$$

The sum is an integer and can be accumulated before one final rational division. For a highest cellwise derivative, $p = 0$, use the same formula away from knots with the positive-part indicator convention. The supplied implementation treats values outside the support as zero before forming the signed sum.

A.3 Reproducing the computation and PDF

From the directory containing the archive contents, run

```
python verify.py
pdflatex Thue_Morse_New_Deductions.tex
pdflatex Thue_Morse_New_Deductions.tex
pdflatex Thue_Morse_New_Deductions.tex
```

The Python script requires Python 3.10 or later and no third-party packages. The LaTeX source uses standard mathematical packages and the New TX fonts available in a full TeX Live installation. The article source is self-contained: it needs no external figures, bibliography database, generated tables, or source-code inclusion to compile.

B Statement ledger

Result	Precisely what is established
Global correction	Theorems 5.1 and 5.3 and corollary 5.2 : all fixed orders, global classical and Hölder norms, explicit constants, and sharp leading supremum asymptotics.
Increasing order	Corollary 5.4 : a supergeometric absolute error certificate with a specified admissible order; no relative endpoint or numerical-conditioning claim.
Dyadic splines	Theorems 6.2 and 6.5 and corollary 6.4 : exact polynomial stabilization, exact degree in lowest terms, finite extraction, and an order-independent absolute amplification bound.
Discrete array	Theorems 7.2 , 8.2 and 9.1 : finite differential conversion, a uniform all-orders local-limit bound, and exact stopping on fixed dyadic odd-level subsequences.
Energy moments	Theorems 10.2 to 10.4 : exact moments, explicit distinct continuous and atomic moment twins, and hence moment indeterminacy.
Geometric extension	Theorem 11.1 : exact separated-copy norms and the resulting moment class for $0 < \theta \leq 1/2$, not for overlapping copies.
Priority and verification	Proofs and finite exact computations are supplied; universal publication priority, peer review, and Lean formalization are not claimed.

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